

Spring REST Exercise 3: REST Custom Exception Handling

Objectives

Demonstrate integration of RESTful Web Service of type GET and test the service using postman.

REST Web Service architecture with Controller, Service and Dao, service methods.

NOTE: There is n- Quiz for this session

Problem Statement - Display Employee List and Edit Employee form using RESTful Web Service

In the previous angular module, we developed a screen that lists employees and it was populated with hard coded values. Now this angular application has be changed t- get the data from RESTful Web Service developed in Spring. The following are the high level activities that needs t- be done t- accomplish this:

Create static employee list data using spring xml configuration

Create a REST Service that reads data from xml configuration and returns it

Make changes in angular component t- consume the created REST Service

Once above activities are completed, clicking on the Edit button against each employee should display Edit Employee form with values retrieved from RESTful Web Service. This will als- involve activities similar t- the one specified above.

NOTE: There is n- specific activity as part of this hands on, refer the next hands ons that covers above three activities in detail.

Create static employee list data using spring xml configuration

Follow steps below t- accomplish this activity:

Incorporate the following in employee.xml:

Create one or tw- more departments

Create four more instances of Employee. (use employee sample data from angular)

Reuse existing skills instead of creating new ones

Include all four employee instances in an ArrayList.

In EmployeeDao, incorporate the following:

Create static variable with name EMPLOYEE_LIST of type ArrayList<Employee>

Include constructor that reads employee list from xml config and set the EMPLOYEE_LIST

Create method getAllEmployees() that returns the EMPLOYEE_LIST

Create REST service t- gets all employees

Follow steps below t- accomplish this activity:

In EmployeeService, incorporate the following:

Change the annotation for this class from @Component t- @Service

Create method getAllEmployees() that invokes employeeDao.getAllEmployees() and return the employee list

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Define @Transactional annotation for this method.

In EmployeeController, incorporate the following:

Include a new get method with name getAllEmployees() that returns the employee list

Mark this method as GetMapping annotation with the URL as '/employees'

Within this method invoke employeeService.getAllEmployees() and return the same.

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Test the service using postman.

Create REST service for department

Create a new service to get all the departments.

Follow steps below to achieve this:

Create a new REST Service, define below list of classes and respective methods:

DepartmentController

getAllDepartments() with URL "/departments", this method will return array of departments

DepartmentService

getAllDepartments()

DepartmentDao

getAllDepartments() - Create a static variable DEPARTMENT_LIST, this should be populated from spring xml configuration

Test the service using postman.

Also verify if department REST service is called by looking into the logs.